Estimating Reclaimed Grass Aboveground Biomass of Reclaimed Mining Area Across Spatial and Temporal Scales With Sentinel-1 and Sentinel-2 Data

Xu Wang, Hang Xu, Fuli Han, Xiaonan Fang, Zhen Zhou, Jianwei Zhou, and Xianhua Yang

***Abstract*—Mining activities destroy vegetation and soil systems, posing a serious threat to ecological security. Reclamation is considered to be an effective and sustainable strategy for improving postmining ecological quality and ecosystem function. Monitoring aboveground biomass (AGB) in reclaimed mining areas is crucial for assessing and managing reclamation. The combination of remote sensing and machine learning algorithms is widely used for regional-scale AGB estimation. However, the feasibility of this approach for assessing AGB in reclaimed mine sites still remains poorly understood. In this study, we employed a variety of data sources (Sentinel-1, Sentinel-2, and their combinations) and coupled them with diverse regression algorithms (including the linear model, LMR; support vector regression (SVR); random forest, RF; and cubist, CB) to estimate herbaceous AGB in typical reclaimed mining areas of Sichuan Province, China. Moreover, we evaluated the temporal–spatial generalizability of the model by training and validating it across distinct geographic regions and time periods. The results showed that 1) with regard to the aspects of model accuracy and stability, the SVR model employing the combination of MSI and SAR data is a superior approach for estimating the AGB of reclaimed mines at the Sentinel scale (RMSE = 90.367 g/m^2, R^2 = 0.597, and mean absolute error (MAE) = 59.721 g/m^2). 2) The most informative SAR metric for estimating AGB is VV. The spectral information from the visible (VIS), red-edge, and shortwave infrared (SWIR) regions is of considerable importance, with significant contributions from both water and soil-adjusted indices. 3) Integration of Sentinel optical and SAR datasets helps to improve AGB estimation for reclaimed mining area grass. 4) The models demonstrated suboptimal performance in temporal and spatial cross-validation (CV), exhibiting limited temporal–spatial generalizability. The proposed framework in this study has great potential to generate robust grass AGB for reclaimed mining areas across regions and time scales. In addition, it provides valuable insights for dynamically guiding the recovery and monitoring of restored ecosystems.**

***Index Terms*—Aboveground biomass (AGB), reclaimed mining area, Sentinel-1, Sentinel-2, temporal–spatial generalizability.**

I. Introduction

MINING plays a significant role in economic and social development, providing employment opportunities, and supplying essential raw materials to society [1]. However, it has a devastating impact on the environment [2]. As a principal method of mining, surface mining significantly alters topography and geological structure by removing overburden and soil through excavation, thereby disrupting surface and groundwater conditions [3]. Surface mining also leads to adverse environmental impacts, including soil erosion, biodiversity loss, and decreased carbon sequestration capacity [3], [4]. Reclamation is significant in the dilemma of economic development and ecological protection [5]. Through reclamation measures such as landform remodeling, soil reconstruction, and revegetation, the structure and functionality of the ecosystem in the mining area can be restored [6]. This is a long-term and dynamic process, requiring continuous monitoring to identify and address issues during reclamation [7].

The vegetation aboveground biomass (AGB) serves as a direct indicator of vegetation quantity and growth status, thereby reflecting the progress of reclamation implementation. Furthermore, it is an important parameter indicating the quality of ecological restoration of mining areas [2], [8]. Remote sensing has been identified as an effective method for quantifying AGB over extensive areas, facilitating precise monitoring at the landscape scale [9]. In recent decades, a large number of methods for estimating AGB have been developed and validated. These methods are mainly based on modeling the relationship between AGB and optical remote sensing reflectance [10], as optical remote sensing provides abundant spectral information and textural features that offer direct and effective support for AGB estimation [11]. In related studies, vegetation indices (VIs) calculated from multiple spectral bands, including the enhanced vegetation index (EVI) and normalized difference vegetation index (NDVI), are widely used as preferred predictors for AGB estimation. This preference stems from their enhanced sensitivity to vegetation biophysical properties [e.g., chlorophyll content, canopy structure, and leaf area index (LAI)] [12]. However, there are certain limitations to the applicability of this method in mining areas where herbaceous is the dominant reclaimed vegetation. The main reason is that, in grassland ecosystems, relatively small canopy

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size and low vegetation coverage lead to a high proportion of soil within each pixel [13]. This causes spatial heterogeneity in soil type, color, and moisture conditions, leading to significant variations in VIs values even at similar AGB levels [14]. This causes spatial heterogeneity in soil type, color, and moisture conditions, leading to significant variations in VIs values even at similar AGB levels [15], overlapping with typical NDVI values for bare soil, which also range between 0 and 0.4 [16]. This soil background effect limits the generalizability of empirical relationships between biomass and VIs, resulting in decreased accuracy of biomass estimation in such areas. To mitigate the impact of soil effects, several soil-adjusted VIs have been proposed, such as the soil-adjusted vegetation index (SAVI) [17], and modified SAVI (MSAVI) [18]. Therefore, incorporating these soil-adjusted VIs when estimating AGB in reclaimed mining areas can improve model robustness and performance. However, there are inherent limitations in estimating AGB using optical remote sensing data [19]. These include: 1) the acquisition of high-quality data is constrained by weather conditions; 2) optical sensors primarily capture information from the upper canopy, providing limited insights into the vegetation structure; and 3) VIs tend to saturate in areas with moderate to high vegetation cover, leading to underestimation of biomass in these regions [20].

In contrast, synthetic aperture radar (SAR) has rapidly become an important data source for vegetation biomass monitoring due to its advantages of all-weather observation, high resolution, and deep penetration [21]. SAR signals can penetrate vegetation canopies and provide detailed information on surface roughness, moisture content, and vegetation structure, which optical sensors cannot fully capture [22]. However, it should be noted that when estimating AGB using SAR data, its signals are susceptible to terrain variations, potentially introducing noise and reducing estimation accuracy [23]. To address these issues, the integration of optical data and SAR data has been proposed to fully harness their complementary advantages [24]. Moreover, the fusion and synergy of multisource data can significantly enhance the retrieval of fine-scale surface information from broad satellite observations [25], thereby improving the accuracy and stability of biomass estimation in grassland ecosystems. Nevertheless, the practical effectiveness of this integration varies, with some studies demonstrating improved pasture grasses AGB estimation accuracy using Sentinel-1, Sentinel-2, and Landsat8 combined datasets, reporting that Sentinel-1 data could effectively improve the estimation accuracy [20]. In contrast, other studies concluded that Sentinel-1 predictors were less effective in estimating grassland biomass across large spatial scales, they also mentioned that the heterogeneity of environmental conditions significantly affects the usefulness of SAR data [26]. As a result, the impact of SAR data on AGB estimation remains unclear. Furthermore, few studies have evaluated the combination of SAR and optical data for estimating AGB in reclaimed mine sites, particularly in reclaimed grassland areas characterized by sparse canopy, mixed soil background, and complex terrain. Therefore, it is critical to explore the integration of different data types in order to develop accurate models for estimating reclaimed grass AGB.

Despite the fact that numerous studies have successfully predicted AGB using remote sensing data, the methods employed may not be effectively generalized to other geographical locations and time periods. This lack of generalizability constrains the broader applicability of remote sensing in predicting AGB [27]. In the case of reclaimed mine areas, the reclamation measures adopted often vary considerably across different sites and time periods, making models developed for a specific site difficult to apply elsewhere. Moreover, the highly fragmented distribution of mines results in disparate environmental conditions, which presents a significant challenge for the implementation of systematic and standardized modeling in large and environmentally heterogeneous regions. Therefore, examining the spatial–temporal generalizability of AGB models is of great practical significance for enabling large-scale and long-term ecological monitoring in reclaimed mining landscapes.

To the best of our knowledge, there are few studies that have tested the combination of synthetic-aperture radar (SAR) and multispectral data to map AGB in reclaimed mine sites. Such maps are useful for the monitoring of ecosystem health and the subsequent management of reclaimed mine sites. The Copernicus program of the European Commission provides open-access, high spatial–temporal resolution optical (Sentinel-2) and SAR (Sentinel-1) imagery, offering promising opportunities for the long-term monitoring of reclaimed mining areas [28]. In particular, Sentinel-2 provides multiple spectral bands useful for vegetation AGB monitoring [29], and Sentinel-1 operates in the short-wavelength C-band, which is highly sensitive to the structural characteristics of herbaceous vegetation and shows strong potential for grassland AGB estimation [30]. Therefore, the objective of this study is to assess the viability of utilizing Sentinel-1 SAR data, Sentinel-2 MSI data, and in situ measurements to estimate grass AGB in reclaimed mining sites. To achieve this goal, we conducted the following: 1) we tested and validated parametric and nonparametric regression algorithms; 2) we detected AGB-related metrics through feature engineering and investigated the effects of data sources, regression algorithms, and their interactions on model performance; and 3) the model performance was further evaluated in terms of robustness, temporal–spatial generalizability by dividing the AGB measurements dataset in different regions and different years.

II. Method

A. Study Area

The study area is located in the Huaying region in the eastern part and the Hongyuan region in the northern part of Sichuan Province, China (see Fig. 1). The Hongyuan area is situated within the humid subtropical climate zone, with an average annual temperature range of 17 °C to 22 °C, and an annual rainfall of 1259.13 mm. The Hongyuan area belongs to the continental monsoon climate zone, with an average annual temperature of 2.9 °C and an annual rainfall of 860.8 mm. The mines in these two areas are dominated by quarries, which over the years have damaged the regional ecosystems. Most of these mines have been abandoned and closed due to the depletion of mine resources and the emergence of geological

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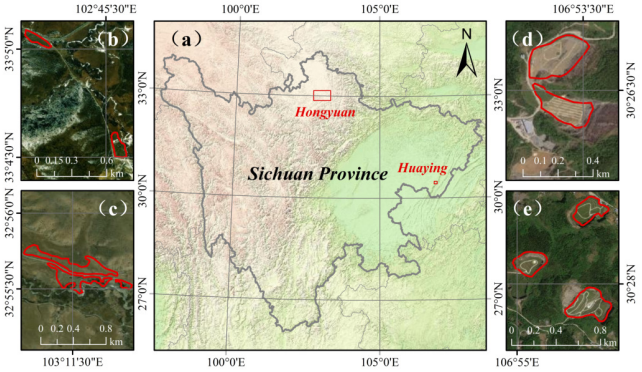


Fig. 1. Location of the study area, (a) location of the Hongyuan and Huaying regions. (b) and (c) Reclaimed mining areas in the Hongyuan region. (d) and (e) Reclaimed mining areas in the Huaying region.

TABLE I
DESCRIPTION OF FIELD SURVEYS AT EACH SITE, AND SUMMARY STATISTICS OF SAMPLING POINTS

Site	Field survey	Number	Min (g/m^2)	Max (g/m^2)	Mean (g/m^2)	S.D. *(g/m^2)
Hongyuan	September 2022	9	104.80	317.00	191.89	72.38
	July 2023	30	4.52	114.04	41.99	29.16
Huaying	September/ October 2022	20	25.50	553.30	255.40	142.14
	October 2023	30	33.20	528.36	178.55	127.37

problems at the mines. Since 2019, Sichuan Province has been carrying out ecological restoration of historical legacy mines through site leveling, soil reconstruction, soil fertilization, and vegetation reconstruction. The reclaimed vegetation is dominated by herbaceous vegetation.

B. In Situ AGB Measurements

The field surveys were conducted at two sites in the study area with the objective of measuring the AGB. A total of 89 in situ AGB sampling plots were collected during the peak vegetation growth period between 2022 and 2023 (July–October). Sampling in 2022 yielded 29 samples due to travel restrictions under China's dynamic zero-COVID policy. To support reliable modeling and analysis, we expanded collections to 60 samples in 2023 following policy relaxation (see Table I).

The sampling plots were 10 × 10 m in extent, consistent with the pixel size of Sentinel-2, and all plots were selected in areas with relatively homogeneous vegetation and soil types. Five quadrats of 0.5 × 0.5 m extent were evenly distributed within a plot, all plants within each quadrat were harvested, and the coordinates of each quadrat was recorded using a global positioning system (GPS) for geographic matching with the Sentinel pixel [see Fig. 2(a)]. AGB was harvested

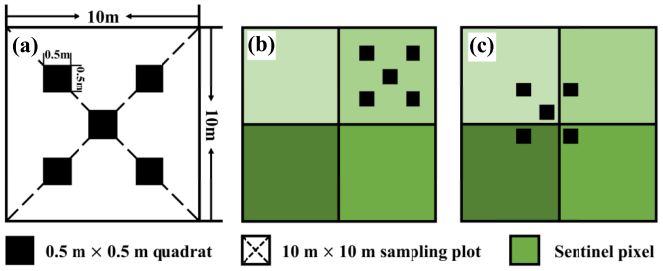


Fig. 2. AGB sampling method and geographic matching strategies with Sentinel pixel. (a) Location of the five quadrats within the sampling plot. (b) All quadrats within a Sentinel pixel. (c) Quadrats not in a Sentinel pixel.

from each quadrat and weighed using an electronic scale. It was then taken back to the laboratory and dried in a drying oven at 75 °C for 48 h. After drying, it was weighed on an electronic scale to obtain the aboveground dry biomass, and the average was used as the AGB for the sampling plot. We used two strategies for the geographic matching of sampling points to Sentinel pixels. First, if all five quadrats within a single Sentinel pixel, the pixel value was directly assigned to represent the sampling point [see Fig. 2(b)]. Second, if some quadrats spanned multiple Sentinel pixels, all pixels containing these quadrats were considered, and the pixel values were calculated by simple averaging to generate the representative pixel value for the sampling plot [see Fig. 2(c)].

C. Sentinel-1 Data and Preprocessing

The Sentinel-1 mission has a constellation of two satellites, the sensors carried by these two satellites are SAR with C-band. Sentinel-1A was launched on April 3, 2014, and Sentinel-1B on April 25, 2016, which exhibit a combined revisit period of 6 days, with a 12-day revisitation rate for each individual satellite. This study used the ground range detected (GRD) images acquired through the interferometric wide (IW) swath mode, which provides dual-polarization imagery (VV and VH). The images have a spatial resolution of approximately 20 × 22 m, with a pixel spacing of 10 m. The Sentinel-1 data were obtained from the European Space Agency (ESA) Copernicus Data Center (https://scihub.copernicus.eu/dhus/#/home) and processed using the SNAP software. The main processing steps include: 1) orbit file apply; 2) thermal noise removal; 3) radiometric calibration; 4) speckle filtering with the refined Lee; 5) range-Doppler terrain correction; and 6) conversion of bands to decibels. After preprocessing, the imagery has a resolution of 10 × 10 m. To ensure consistency with the field data, Sentinel-1 images were selected based on acquisition dates that were temporally closest to the corresponding field survey campaigns. The Sentinel-1 images were selected. For the Hongyuan region, the data were acquired on September 26, 2022 and August 4, 2023, for the Huaying region, on September 28, 2022 and October 5, 2023.

D. Sentinel-2 Data and Preprocessing

The Sentinel-2 mission has a constellation of two satellites in the same orbit. The sensors carried by these two satellites are Multispectral Instruments (MSI). Sentinel-2A was

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TABLE II
METRICS CALCULATED FROM SENTINEL-1 DATA

Parameters	Equation
Polarization multiplicative	VV × VH
Simple polarization ratio	VV−VH, VH−VV
Polarization division	VV/VH, VH/VV
Polarization averaging	(VV + VH)/2
Polarization multiplicative averaging	(VV × VH)/2

launched on June 23, 2015 and Sentinel-2B on March 7, 2017, which exhibit a combined revisit period of 5 days, with a 10-day revisitation rate for each individual satellite. The MSI comprises 13 spectral bands and three spatial resolutions of 10 m (blue, green, red, near-infrared (NIR) bands), 20 m (vegetation red edge bands, narrow NIR band, and shortwave-infrared (SWIR) bands), and 60 m (coastal aerosol, water vapor, SWIR-cirrus bands). This study used the Sentinel-2 Level-2A data, which were obtained from the ESA Copernicus Data Center. The Sentinel-2 L2A products are processed with Sen2Cor, which uses Level 1C top of atmosphere (TOA) inputs, performs atmospheric, terrain, and cirrus corrections, and generates bottom of atmosphere (BOA) reflectance multiplied by 10 000 as the final output. In our study, we used the bands with spatial resolutions of 10 and 20 m, with the 20-m bands subsequently resampled to 10 m using the SNAP software. Sentinel-2 images were selected based on acquisition dates closest to the field surveys and with minimal cloud cover. For the Hongyuan region, the data were acquired on September 11, 2022 and July 8, 2023, for the Huaying region, on September 30, 2022 and September 5, 2023.

E. Extraction of Predictors

Forty seven metrics were selected from Sentinel-1 SAR and Sentinel-2 MSI data as predictors of AGB. For Sentinel-1 data, the backscatter values of both VV and VH polarizations were selected for estimating AGB. In addition, several metrics have been proposed as potential predictors of AGB (see Table II) [31], which have also been included in the predictor set.

For Sentinel-2 data, ten of the thirteen bands from S2 (four visible (VIS), four red-edge, and two short-wave infrared bands), were selected for creating the predictor set. Previous studies have shown that numerous spectral vegetation index provide a richer source of information than individual spectral bands for retrieving AGB [32]. The 25 VIs calculated by the S2 bands were selected for analysis in this study (see Table III). The biophysical parameters have also been demonstrated to be effective in estimating AGB [33]. The LAI, fraction of vegetation cover (fCover), and fraction of absorbed (fPAR) were calculated using the biophysical processor of SNAP. The processor employs a neural network algorithm based on the PROSAIL radiative transfer model [34].

F. Machine Learning Regressions

Four regression algorithms were considered in this study, including two types of algorithms: 1) parametric methods

TABLE III
VEGETATION INDEX CALCULATED FROM SENTINEL-2 DATA

Vegetation Index	Abbr.	Equation	Reference
Difference Vegetation Index	DVI	DVI = B8 − B4	[35]
Enhanced Vegetation Index	EVI	EVI = 2.5 × ((B8 − B4)/(B8 + 6 × B4 − 7.5 × B3))	[36]
Red-edge chlorophyll index	CIre	CIre = B8/B6 − 1	[37]
Green Normalized Difference Vegetation Index	GNDVI	GNDVI = (B8 − B3)/(B8 + B3)	[38]
Green-Red Vegetation Index	GRVI	GRVI = (B3 − B4)/(B3 + B4)	[39]
Infrared Simple Ratio 1	ISR1	ISR1 = B8/B11	[40]
Infrared Simple Ratio 2	ISR2	ISR2 = B8/B12	[40]
Land surface water index	LSWI	LSWI = (B8 − B11)/(B8 + B12)	[41]
Modified Soil-Adjusted Vegetation Index	MSAVI	MSAVI=(1 − B8) − 1/2 [(2 × B8+1)^2 − 8(B8 − B4)]$^{1/2}$	[18]
Normalized difference phenology index	NDPI	NDPI = (B8 − (0.74 × B4 + 0.26 × B11))/(B8 + (0.74 × B4 + 0.26 × B11))	[42]
Normalized Difference Vegetation Index	NDVI	NDVI = (B8 − B4)/(B8 + B4)	[43]
Kernel Normalized Difference Vegetation Index	kNDVI	tanh(NDVI^2)	[44]
Normalized Difference Water Index 1	NDWI1	NDWI1 = (B8 − B11)/(B8 + B11)	[45]
Normalized Difference Water Index 2	NDWI2	NDWI2 = (B8 − B12)/(B8 + B12)	[46]
Normalized Difference Water Index 3	NDWI3	NDWI3 = (B3 − B8)/(B3 + B8)	[47]

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TABLE III
(Continued.) VEGETATION INDEX CALCULATED FROM SENTINEL-2 DATA

Red edge normalized difference vegetation index 1	RENDVI1	RENDVI1 = (B8 − B5)/(B8 + B5)	[48]
Red edge normalized difference vegetation index 2	RENDVI2	RENDVI2 = (B8 − B6)/(B8 + B6)	[48]
Red-Edge Normalized Difference Water Index 1	RENDWI1	RENDWI1 = ((B5 − B11))/((B5 + B11))	[49]
Red-Edge Normalized Difference Water Index 2	RENDWI2	RENDWI2 = (B6 − B11)/(B6 + B11)	[49]
Soil-Adjusted Vegetation Index	SAVI	SAVI = 1.5 × (B8 − B4)/(B8 + B4 + 0.5)	[17]
Simple Ratio	SR	SR = B8/B4	[50]
Modified Normalized Difference Vegetation Index	MNDVI	MNDVI = (B8A − B12)/(B8A + B12)	[51]
Chlorophyll red-edge	CHLRE	CHLRE = $(B7/B5)^{-1}$	[52]
Vogelmann Red Edge Index 1	VREI1	VREI1 = B8/B5	[53]
Vogelmann Red Edge Index 2	VREI2	VREI2 = B8/B6	[53]

(LMRs) and 2) nonparametric methods [support vector regression (SVR), random forest (RF), and cubist (CB)]. All algorithms were implemented in the CRAN R package caret (see Table IV). The use of linear regression models to estimate AGB using remote sensing data is commonly used because of their simplicity and interpretability [54]. However, traditional linear models (LMRs) are affected by the issue of multicollinearity [55]. Regularization methods, such as ridge regression, mitigate the impact of redundant variables by shrinking their coefficients, which is helpful in solving this issue [56]. Therefore, LMR was selected as the baseline model for reference. Given that biomass is usually nonlinearly correlated with remotely sensed variables, nonparametric models (such as SVR, RF, and CB) have been widely used for satellite-based AGB estimation. This is done to improve the nonlinear estimation of biomass models [57]. Such models do not make a priori assumptions about the probability distributions of the variables, and therefore have significant advantages over parametric statistical models that assume Gaussian distributions [32]. In addition, nonparametric machine learning techniques are capable of processing high-dimensional features and identifying complex nonlinear relationships between variables in biomass modeling [58]. Such techniques provide more accurate estimates of AGB than linear regression models, especially when using data from multiple sources [9]. Specifically, SVR demonstrates strong robustness to outliers in high-dimensional data and exhibits good generalization ability, particularly with small sample sizes [59]. RF through its ensemble learning mechanism, shows strong resistance to multicollinearity, data noise, outliers, and overfitting [60]. CB based on a gradient boosting framework, enables efficient training across different regions and spatial scales and significantly improves prediction accuracy [61].

G. AGB Modeling Framework

We divided the Sentinel dataset into three AGB predictor sets: 1) nine Sentinel-1SAR metrics; 2) 38 Sentinel-2 MSI metrics; and 3) combination of Sentinel-1 and Sentinel-2 metrics (47 metrics). All metrics were normalized [centered on the mean and scaled by the standard deviation (SD)]. We checked the linear dependencies of each metrics using the variation inflation factor (VIF). A high threshold (VIF >10) was adopted to remove variables with high collinearity for avoiding severe multicollinearity among the selected predictor variables. The four regression algorithms (LMR, SVR, RF, and CB) were applied to the three predictor sets. The optimal model parameters were determined using the grid search method, and the parameters with the lowest root mean squared error (RMSE) were selected through fourfold cross-validation (CV).

In the process of AGB modeling, one of the challenges is to identify the most informative metrics set that improves the predictive power of the AGB and helps to interpret the final model, while also ensuring that the set is not too large and therefore not overly burdensome to compute. We applied the recursive feature elimination (RFE) algorithm for feature selection. The RFE was used to evaluate the impact of the number of input features on the model performance, which was determined by the RMSE and quantified in a fourfold CV process that was repeated ten times. The RFE algorithm fits a model to all predictors, and the predictor were ranked according to their importance to the model. The variable set is divided into n variable subsets according to the importance, with n representing the number of variables. For each subset of variables (S_i, i = 1, 2, 3,.,n), the top i most important variables are selected to construct a new dataset. The model is then refitted, and the performance is evaluated based on the new dataset. In this study, we chose the variable subset with the lowest RMSE as the optimal variable subset.

When the available datasets are limited, making validation with a separate dataset infeasible, K-fold CV offers an economical and effective solution, as it allows the use of all available data for both training and validation. However,

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TABLE IV
DESCRIPTION OF THE REGRESSION ALGORITHM, INCLUDING THE PARAMETERS CONSIDERED, THE BASIS FOR FEATURE RANKING, AND THE REQUIRED R PACKAGES

Type	Model	Abbr.	Parameters	Feature rank criteria	R package
Parametric Model	Linear Model with Ridge Regularization	LMR	$lambad$ = 0.01, 0.5, 1	Absolute value of coefficients	glmnet, Matrix
Non-parametric Model	Support Vector Machines with Radial Basis Function Kernel	SVR	Cost = 10^n(n = -5, -4, -3, ...0, 1, 2) Singma = 0.1, 0.2, 0.3, ...,1	Combination of the absolute values of the weights*	kernlab
	Random Forest	RF	n.tree = 300, 500, 700, 900, 1000, 1500 mtry = $\sqrt{k}$ (k is the number of variables entered)	Decrease in accuracy by permuting a variable	randomForest
	Cubist	CB	committees = 10,20,30 neighbors = 9	linear combination of the usage in the rule conditions and the model	Cubist

*Achieved through vip package.

CV may produce optimistically biased performance estimates [62]. To reduce this potential bias and obtain a more robust assessment of model generalization performance, we followed the recommendation of repeating the CV ten times [63]. Therefore, in this study, we used a fourfold CV repeated ten times to evaluate model performance, using the coefficient of determination (R^2), RMSE, mean absolute error (MAE), and bias as performance metrics. Furthermore, to assess the variability in model predictions and quantify the uncertainty in AGB estimates, the CV procedure was repeated 100 times. The mean and SD of the AGB estimates for each pixel were then calculated across these 100 runs, with the mean representing the final predicted AGB and the SD indicating the associated uncertainty (see Fig. 3).

Two-way analysis of variance (ANOVA) was used to assess the effects of data source, regression method, and their interaction on model performance metrics (RMSE, R^2, and MAE). Subsequently, Tukey's honestly significant difference (HSD) test was conducted to compare model accuracy across different data sources.

H. Temporal and Spatial Generalizability

We conducted temporal and spatial CV tests for exploring the temporal–spatial generalizability of the model. In the spatial CV tests, the sampling plots in the Hongyuan (39 samples) and Huaying (50 samples) reclaimed mining areas were selected for model development, respectively, with one dataset used for training and the other dataset used for validation. In the temporal CV tests, the model was trained using the 29 sampling plots in 2022, and validated using the 60 sampling plots in 2023. In the other model, the training and validation sets were switched.

III. RESULTS

A. Selection of Optimal SAR and MSI Metrics to Estimate AGB

The specific optimal metrics subset was selected for each dataset and regression method by using the RFE algorithm (see Fig. 4). For example, models using only SAR data selected between 8 and 9 indicators (9 for LMR, 8 for SVR, 8 for RF, and 9 for CB); models using only MSI data selected between 5 and 13 (13 for LMR, 9 for SVR, 5 for RF, and 13 for CB); and models using combined data selected between 14 and 31 indicators (16 for LMR, 18 for SVR, 14 for RF, and 31 for CB). LMR solves the limitation of traditional LMRs in handling high-dimensional data by regularization, and the increase in the number of variables had less impact on the RMSE when the model performance reaching the optimum. In nonparametric models, CB is significantly affected by the number of input indicators, and its performance tends to fluctuate within a certain range depending on the number of

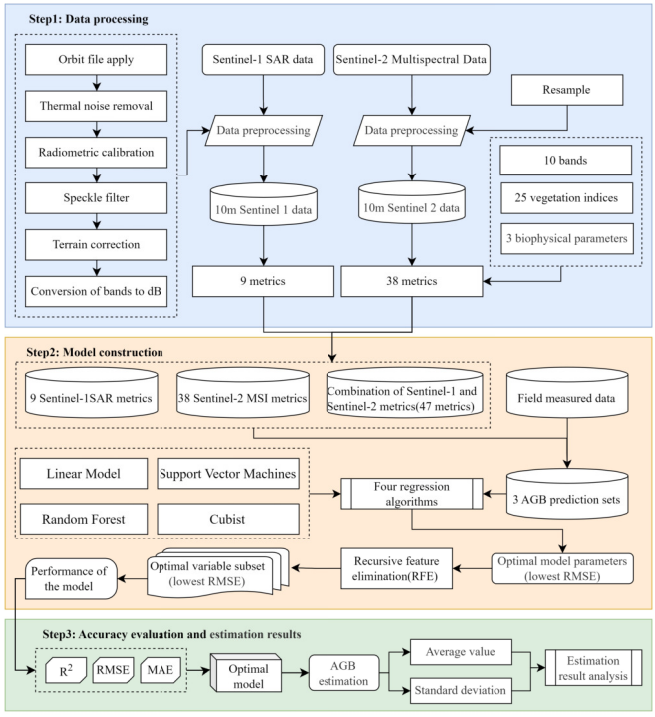


Fig. 3. AGB modeling framework.

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TABLE V

ANOVA AND TUKEY'S HSD TESTS WERE CONDUCTED ON RMSE, R^2, AND MAE TO EVALUATE THE EFFECTS OF DATA SOURCE, REGRESSION METHOD, AND THEIR INTERACTION

Metric	ANOVA			Tukey's HSD tests		
	Factor	F value	p-value	Factor	difference	Adjusted p-value
RMSE	Data	26281	$<2e^{-16}$	SAR-SAR+MSI	17.392	$<1e^{-16}$
	Method	10191	$<2e^{-16}$	MSI-SAR+MSI	2.499	$1.7e^{-6}$
	Data: Method	2037	$<2e^{-16}$	SAR-MSI	14.893	$<1e^{-16}$
R^2	Data	34354	$<2e^{-16}$	SAR-SAR+MSI	-0.222	$<1e^{-16}$
	Method	10971	$<2e^{-16}$	MSI-SAR+MSI	-0.030	$6e^{-7}$
	Data: Method	2110	$<2e^{-16}$	SAR-MSI	0.192	$<1e^{-16}$
MAE	Data	45508	$<2e^{-16}$	SAR-SAR+MSI	16.782	$<1e^{-16}$
	Method	21237	$<2e^{-16}$	MSI-SAR+MSI	0.853	0.257
	Data: Method	4108	$<2e^{-16}$	SAR-MSI	15.930	$<1e^{-16}$

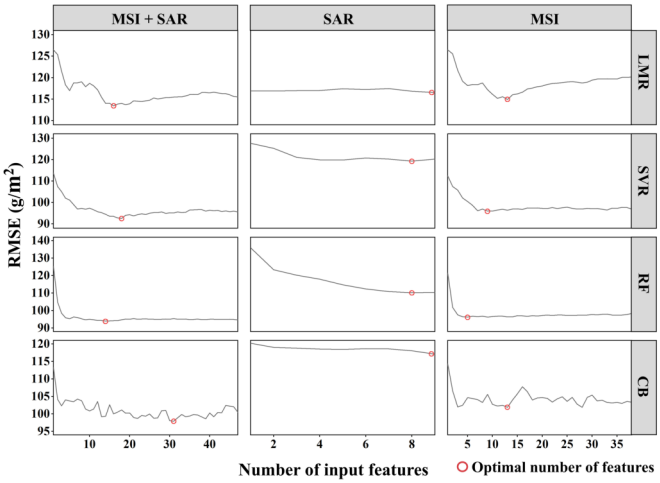


Fig. 4. Effect of the subset feature size on the RMSE for the regression methods and data sources used, red hollow circle was the best size of features with the lowest RMSE.

input indicators. SVR and RF also have robust capabilities in handling high-dimensional features. They are less affected by the number of input indicators. Furthermore, the addition of more indicators after the model has reached its optimal performance does not significantly affect the model's overall performance.

The contribution of each data source to the final model, including the number of selected indicators and their importance to the model performance, depended on the regression method used (see Fig. 5). When using a combined dataset of SAR and MSI data, all models selected both SAR and MSI indicators, with more MSI indicators chosen than SAR indicators. Specifically, LMR selected 3 SAR indicators and 13 MSI indicators, with MSI indicators playing a significant role in the model performance. The SVR model selected 5 SAR indicators and 13 MSI indicators, with the SAR indicators being significantly important to the model. RF selected 2 SAR indicators and 12 MSI indicators, demonstrating a more balanced contribution to the model performance. CB selected 7 SAR indicators and 24 MSI indicators, making it the model with the highest number of SAR indicators selected.

B. Performance of Models for AGB Modeling

The distribution of model performance for different classification algorithms with different data sources was derived by running the CV program 100 times (see Fig. 6). For detailed statistical information on model performance, please refer to the supplementary files (see Table S1). Furthermore, the ANOVA revealed that model performance differed significantly depending on the dataset, the classification algorithm, and their interaction (p-value <0.001) (see Table V). For detailed information on ANOVA and Tukey's HSD tests, please refer to the supplementary files (see Tables S2 and S3). The RMSE of the models ranged from 87.866 to 118.871 g/m^2, the R^2 ranged from 0.240 to 0.626, and the MAE ranged from 58.017 to 89.636 g/m^2. Considering these three metrics and the results of Tukey's HSD test, the model with the highest accuracy is the SVR model using the combination of MSI and SAR data, which performed significantly better than LMR and CB (adjusted p-value less than 0.05), and slightly better than RF (see Table S3). Specifically, its RMSE ranging from 88.360 to 92.305 g/m^2 with a mean of 90.367 g/m^2, R^2 ranging between 0.570 and 0.626 with a mean of 0.597, and MAE ranging from 58.017 to 61.088 g/m^2 with a mean of 59.721 g/m^2.

According to the results of Tukey's HSD tests (see Table V), the integration of SAR and MSI data yielded significant enhancements in model performance across all regression methods except LMR. Specifically, RMSE and MAE were significantly reduced (i.e., negative differences with adjusted p-values less than 0.05), while R^2 was significantly increased (i.e., positive differences with adjusted p-values less than 0.05). Compared to models using single SAR datasets, the integrated SAR-MSI datasets demonstrated marked performance enhancement, with mean RMSE and MAE decreased by 4.541–26.105 g/m^2 and 1.051–24.406 g/m^2, and R^2 increased by 0.069–0.311. Compared to models using single MSI data, the performance of models using combined datasets also yield consistent gains: mean RMSE and MAE decreased by 1.392–3.779 g/m^2 and 0.681–3.488 g/m^2, and R^2 improved by 0.019–0.036. Crucially, MSI data contributed predominantly to the performance boost, compared to SAR data.

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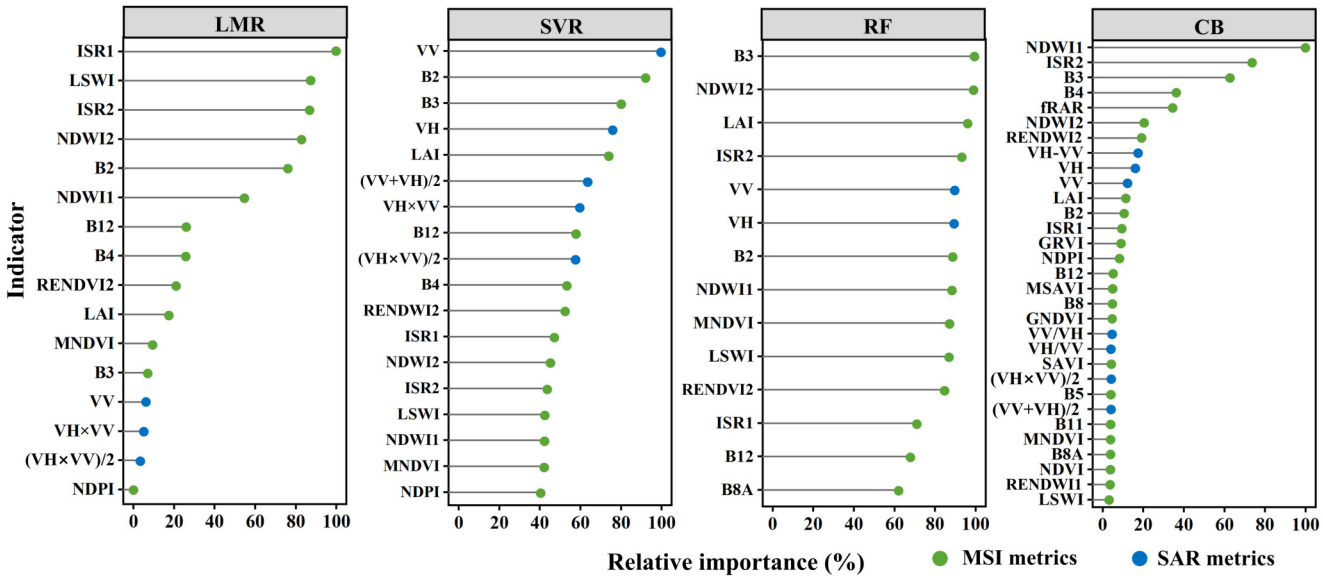


Fig. 5. Relative importance of the indicator for each regression method with the combined SAR and MSI dataset.

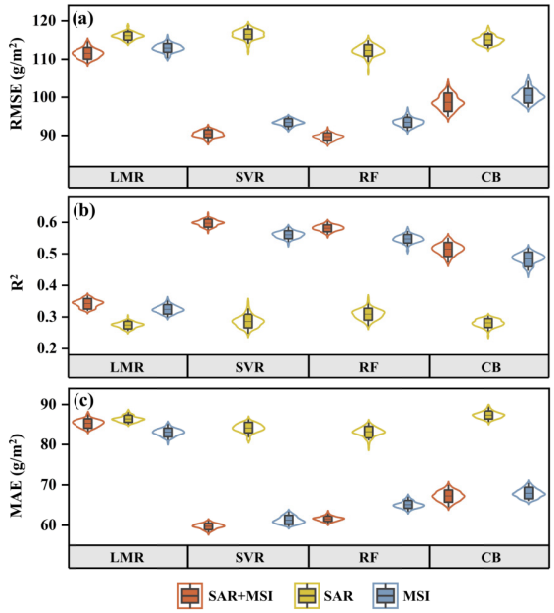


Fig. 6. Distribution of the 100-CV (a) RMSE, (b) R^2, and (c) MAE for each regression method and data source.

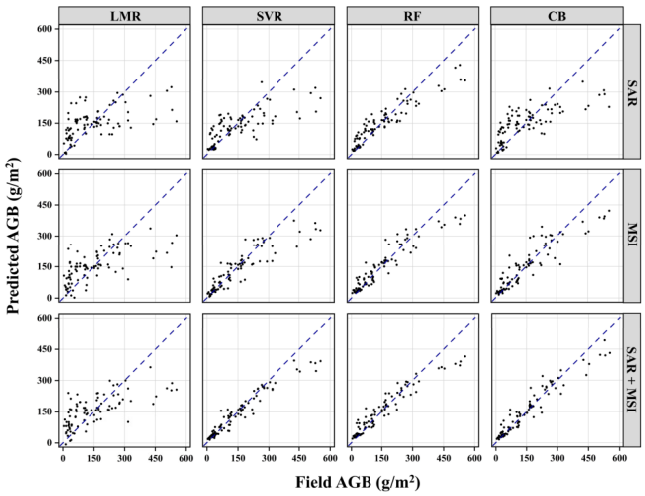


Fig. 7. Comparison of field AGB with predicted AGB (mean of CV predictions from 100 model runs) from different methods and data sources.

Therefore, the performance enhancement achieved through data fusion is primarily driven by the MSI data. In addition, among the relatively best performing SVR and RF models, using combined datasets improved model stability compared to using single datasets. This is reflected in the reduced variability of model performance metrics, as evidenced by a decrease in SD. The SD of RMSE decreased by 0.274–0.365, the SD of MAE decreased by 0.224–0.543, and the SD of R^2 decreased by 0.003–0.008.

Comparison of the average AGB predicted by running the CV procedure 100 times for different models with field AGB (see Fig. 7). All models demonstrated a tendency to overestimated AGB for values <150 g/m^2 and underestimated AGB for values >300 g/m^2. This tendency toward overestimation at low biomass values and underestimation at high biomass values was particularly pronounced, primarily due to the saturation effect of VIs. Notably, single-dataset models (particularly SAR-only) exhibited pronounced instability, which was effectively attenuated through integrated SAR-MSI datasets.

C. Spatial Distribution of AGB

The best models for biomass estimation (SVR based on SAR and MSI metrics) were used to map the spatial distribution of the AGB estimations across the study area. The mean of the AGB estimates for each pixel was used as the final AGB prediction (see Fig. 8), while the SD was employed to represent the uncertainty in the AGB prediction (see Fig. 9). The spatial distribution of predicted AGB exhibited large spatial variability both within and between reclaimed mining sites. In the three reclaimed mining sites within the Hongyuan region, the estimated AGB ranged from 19.967 to

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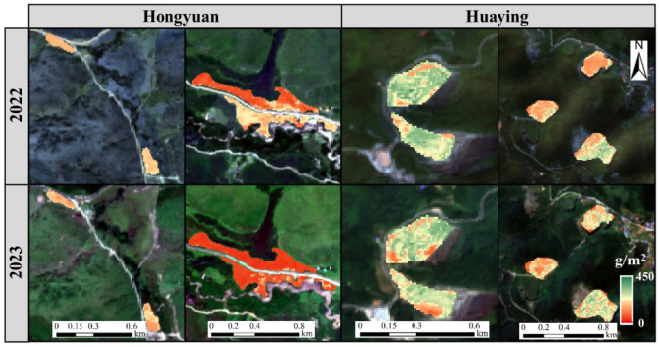


Fig. 8. Spatial distribution of mean AGB based on 100 runs of CV procedure.

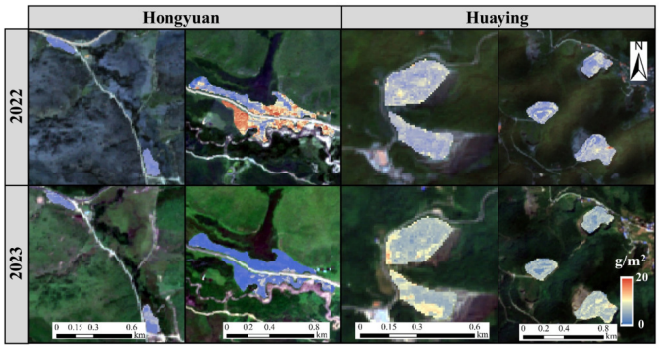


Fig. 9. Uncertainty of the AGB (reported as SD) based on 100 runs of CV procedure.

292.520 g/m^2 with an SD from 0.614 to 12.731 g/m^2. In the five reclaimed mining sites selected within the Huaying region, the estimated AGB ranged from 29.111 to 411.538 g/m^2 with SDs between 0.896 and 18.722 g/m^2. The AGB SD was greater at the locations with very low or high predicted AGB. The predicted AGB for most regions exhibited less variability between years, while in the southern part of the Hongyuan Mining sites, the rapid turnover of vegetation in reclaimed mine sites resulted in significant interannual variations in predicted AGB.

D. Temporal–Spatial Generalizability of the Models

To assess the temporal generalizability of the AGB regression models, 29 sampling plots in 2022, and the 60 sampling plots in 2023 were used to build the models, respectively. In the model evaluation, the dataset used is different years than the dataset from which the model was constructed. To assess the spatial generalizability of the AGB regression models, the sampling plots in the Hongyuan (39 samples) and Huaying (50 samples) reclaimed mining areas were selected for model development, respectively, with one dataset used for training and the other dataset used for validation. The models employed are four regression algorithms using the optimal dataset (SAR and MSI) (see Table VI). None of the four models has good temporal and spatial generalizability at the same time. In the temporal CV, when predicting AGB for 2023 using the model established with sample plots from 2022, the RF model had the highest R^2 (0.239) among the four regression models. Its RMSE and MAE were also lower compared to the other models. Conversely, when predicting AGB for 2022 using the regression model established with sample plots from 2023, the CB model had the highest R^2 (0.139) among the four regression models. Its RMSE and MAE values were also lower compared to the other models. All the models had low R^2 and high RMSE and MAE, indicating that none of them have good temporal generalizability. In the spatial CV, when using the regression model established with sample plots from the Huaying area to predict the vegetation biomass in the Hongyuan area, the LMR model had the lowest R^2 (0.016) among the four regression models. The R^2 for the SVR, RF, and CB models were 0.398, 0.239, and 0.198, respectively. When the training set and validation set were swapped, the CB model had the highest R^2 (0.435) among the four regression models, and its RMSE and MAE values were also lower compared to the other models. In summary, none of these four regression models demonstrated good spatial generalizability with this dataset. Therefore, in future AGB modeling work, it is necessary to continue collecting field data and to take into account factors such as environment, topography, and geographic location during model training to achieve high-accuracy prediction results.

IV. Discussion

A. Relationship of S1 SAR and S2 MSI With AGB Modeling

This study evaluated the capabilities of multisource variables derived from MSI and SAR for modeling reclaimed vegetation AGB. For optical data, the information from the VIS, NIR, and SWIR presents the most important capabilities in AGB modeling [64]. In previous studies, the NDVI and EVI constructed using NIR bands are still the two most widely used vegetation indexes [65]. However, in our results, these indicators were only selected by the CB, and their importance was also relatively low. Indeed, the importance of using SWIR-based vegetation indexes in combination with individual spectral bands was clearly demonstrated in our model selection results. For instance, in the single band selected for all four models, the VIS bands (B2, B3, and B4) related to vegetation pigmentation play a significant role in the model. The same was confirmed for the SWIR2 band, which is associated with nitrogen, lignin, and cellulose [66]. The vegetation indexes selected for all four models demonstrated the greatest performance among those including red-edge and SWIR band (i.e., NDWI and RENDWI), which account for the water content of the vegetation. Previous studies have shown a correlation between these biochemical traits and canopy structure [67]. Therefore, estimating the canopy structure (e.g., AGB) of reclaimed vegetation should consider obtaining additional information from sources other than the NIR, including VIS, red-edge, and SWIR.

In terms of the radar polarimetric parameters, the AGB model relying solely on the SAR dataset underperformed, largely due to the complex topographic in the mining area. First, the C-band SAR exhibits higher sensitivity to surface moisture than biomass, particularly in areas with sparse vegetation [28]. Second, AGB estimation using SAR data is subject to greater topographic influence [23], especially in mining areas with greater topographic variability, and this adverse effect may be amplified in rugged terrains. Recent

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TABLE VI
TEMPORAL–SPATIAL GENERALIZABILITY PERFORMANCE COMPARISON OF AGB REGRESSION MODELS BY USING DIFFERENT DATASET

Model	Temporal-spatial generalizability	Training dataset	Validation dataset	RMSE (g/m^2)	R^2	MAE (g/m^2)
LMR using SAR and MSI	Temporal generalizability	2022	2023	203.282	0.065	178.370
		2023	2022	187.497	0.079	146.718
	Spatial generalizability	Huaying	Hongyuan	190.094	0.398	171.817
		Hongyuan	Huaying	198.195	0.185	153.202
SVR using SAR and MSI	Temporal generalizability	2022	2023	219.668	0.152	183.186
		2023	2022	182.292	0.087	159.406
	Spatial generalizability	Huaying	Hongyuan	179.627	0.016	158.291
		Hongyuan	Huaying	220.400	0.134	177.586
RF using SAR and MSI	Temporal generalizability	2022	2023	192.758	0.239	167.639
		2023	2022	137.201	0.095	106.606
	Spatial generalizability	Huaying	Hongyuan	199.179	0.413	183.377
		Hongyuan	Huaying	207.107	0.225	166.512
CB using SAR and MSI	Temporal generalizability	2022	2023	191.721	0.053	169.856
		2023	2022	171.336	0.139	136.793
	Spatial generalizability	Huaying	Hongyuan	211.705	0.198	171.268
		Hongyuan	Huaying	215.481	0.435	199.374

studies on grassland monitoring have confirmed that the integration of optical and SAR datasets improves herbaceous AGB prediction capacities through complementing each other and reducing negative influences of soil background and weather conditions [24]. Similarly, this has also been confirmed in our study, where the accuracy of the model improved after integrating the S1 and S2 datasets. This is attributed to the complementary information provided by SAR and MSI. Based on the vegetation surface reflectance features provided by MSI, while SAR captures additional information on the structure of the vegetation [22], [68]. For instance, in this study, VV and VH polarizations showed similar importance in grassland AGB estimation models. This may be because VV polarization captures the vertical structure of vegetation, while VH polarization is more sensitive to multidirectional scattering from leaves and stems, thus providing complementary detailed information [69]. The combination of the two offers a comprehensive representation of the structural and scattering characteristics of herbaceous vegetation. Such complementary information is particularly valuable for reclaimed herbaceous vegetation AGB estimation, helping to enhance model performance and mitigate the bias in estimating low and high AGB values.

B. Accuracy Assessment of the Reclaimed Mining Area AGB Models

Our modeling approach provided effective multitemporal estimates of herbaceous biomass at the maturity stage for reclaimed vegetation. The performance of the four models used (LMR, SVR, RF, and CB) varied considerably when predicting across a wide range of conditions. The optimal models (SVR model using the combination of MSI and SAR data) had low error (RMSE of 16.653 to 17.650 g/m^2) and good model fit (R^2 of 0.240 and 0.626). Influencing the performance of the AGB model is explained by several factors, including: 1) the regression technique chosen for analysis; 2) the number and type of metrics selected as potential input data; 3) the type of vegetation studied; and 4) the quality of the field and remote sensing data used to obtain the models.

The results of the machine learning algorithms show that there is potential for improvement in the estimation capabilities of AGB. Compared to the nonparametric algorithm, the parametric algorithm incorporating regularization does not produce satisfactory results when faced with high-dimensional datasets. This confirms that the enhanced flexibility of nonparametric algorithms in the use of variables and in dealing with complex relationships depends on the fact that they impose fewer constraints on the relationship between the independent and dependent variables [70]. Therefore, more complex nonparametric models have advantages in dealing with the interrelationship between vegetation biophysical properties and complex remote sensing data [71]. Especially, SVR uses kernel methods to more effectively model complex nonlinear patterns, making it particularly suitable for small sample datasets, while also controlling model complexity through regularization mechanisms (i.e., penalty parameters and loss functions), thereby enhancing generalization

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ability [31]. Therefore, SVR holds strong potential for AGB estimation.

Choosing the most appropriate metrics to estimate AGB is another factor that may affect the performance of the model. The feature optimization procedure is essential when using a large number of training features [72]. This is confirmed by our study, the number of metrics used as inputs can be greatly reduced without decreasing the accuracy of the AGB estimates. Even regression methods that are incompletely affected by high data dimensionality can benefit from a feature selection procedure. More detailed feature selection procedures can generate streamlined models for practical applications and improve the operational efficiency of the models. Therefore, models based on complex multisource datasets need to tradeoff high data dimensionality and information loss to obtain an appropriate number of features.

The complexity of vegetation in the study area also affects the accuracy of AGB estimates. Several reclaimed mines are covered in our study area, some of which are even far away from each other. Differences in reclamation measures and subsequent management tools in different mines led to differences in vegetation types between mines, as well as changes in vegetation types at different time periods in the same mine due to vegetation succession, all of which affected the accuracy of the model [2]. Hongyuan mining area pays attention to the restoration of damaged vegetation and the rational development of the mining area, while Huaying mining area pays attention to adding new vegetation planting areas and stopping the exploitation of resources. This difference in development and restoration may also be the reason for the difference in estimation performance between the two areas. Species composition information has been found to improve AGB estimation in previous studies [73], in which compositional heterogeneity is strongly related to variability in herbaceous biomass production potential [64]. Therefore, we believe that in reclaimed mine areas with complex vegetation types and varying restoration approaches, incorporating vegetation composition information into AGB estimation models may be more critical, as it helps to more accurately capture the spatial heterogeneity of AGB.

Modeling AGB in diverse and topographically complex reclaimed mining areas has some obstacles, such as the acquisition of high-quality and standardized field data and the integration of field data with remotely sensed data. 1) The number and representativeness of ground sampling points. The collection of a sufficient number of representative biomass sample plots is often a constraint to the development of robust AGB estimation models, as it is both time-consuming and labor-intensive. In addition, the closed management practices of some reclaimed mining areas further hinder access to adequate samples. In this study, we adopted ten-fold repeated CV to make full use of the limited available samples and obtain more robust model evaluation results. In future research, generative adversarial networks can be introduced to expand the training dataset, thereby enhancing the generalization ability of models under small-sample conditions while preserving data diversity. Moreover, employing UAV-based remote sensing to explore a nondestructive approach for estimating grassland AGB could enable rapid and efficient acquisition of samples, further improving model accuracy and applicability. 2) The degree of correspondence between sampling points and satellite data. The degree of correspondence between sampling points and satellite data. Temporal matching between most in situ AGB sampling and satellite imagery is constrained by severe weather limitations affecting optical remote sensing [74], and cloud contamination further degrades the quality of remote sensing data. In the future, the use of cloud-free composite images could be considered to mitigate this issue. At the same time, the impact of physiological changes during different growth stages of grassland on remote sensing inversion models should be fully considered. One effective approach is to introduce time-series analysis methods for continuous prediction of biomass dynamics [75], to avoid the negative effects of seasonal variability on the model. In terms of spatial matching, the complex and fragmented patches of reclaimed mines may result in mixed pixels [26] and edge effects [71]. In our study, most of the sampling points were selected to be pure vegetation within a space of at least 10×10 m (a Sentinel pixel) to minimize the negative effects. Moreover, scale effects between in situ data and remote sensing data [76] represent another significant source of uncertainty in AGB modeling [77]. Therefore, a series of 0.5×0.5 m quadrats was set up, and their calculated averages were aligned with the pixels in the study to reduce the adverse impacts of scale effects and spatial misalignment, though some uncertainty may still remain due to spatial resolution and GPS positioning errors.

C. Temporal–Spatial Generalizability of the Models

Assessing the temporal–spatial generalizability of AGB models is of great significance for enabling large-scale or long-term monitoring in reclaimed mining areas. Such assessment facilitates more scientific planning and management of reclamation projects by decision-makers, while also optimizing the time and cost efficiency of monitoring efforts. In our evaluation of the temporal–spatial generalizability of the model, the generalizability of AGB models trained with data from a reclaimed mining area or one year and tested on another mine or another year was found to be limited. This phenomenon can be contributed to the classical problem of domain shift in machine learning, where the feature space of the test data and the training data differ significantly [26]. The negative impacts of this may be amplified when estimating AGB for reclaimed vegetation. The diversity of plant species and growth on reclaimed mine sites varies in both space and time, as it depends on three primary factors: 1) relatively variable abiotic environmental conditions (such as climate, topography, and soil texture); 2) more variable abiotic conditions (such as weather and precipitation patterns); and 3) land use activities (such as management activities after reclamation). This leads to large differences in characteristics between mine sites and over time. In addition, machine learning algorithms themselves may also have limitations. Nonparametric models primarily rely on training samples for learning, particularly local information [78]. They assume no specific form of data distribution, and can therefore capture complex data structures and patterns.

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However, this flexibility comes at the cost of limited predictive power when dealing with data beyond the sample range [70]. Therefore, the supplementation of corresponding ground truth measurements is crucial when applying remote sensing-driven predictive models to estimate AGB in a new year or at a new reclaimed mining site. In addition, enhancing the temporal–spatial generalizability of AGB models should be a key focus of future research. This could involve integrating multisource remote sensing data to improve the robustness of feature representation, and introducing domain adaptation or transfer learning techniques to reduce the distribution discrepancy between the source and target domains, thereby alleviating the problem of limited training samples in the target domain [79]. By improving the temporal–spatial generalizability of models, it becomes possible to apply them effectively across broader reclaimed areas and over different time periods, while minimizing the need for additional ground-based data collection. This would significantly enhance the scalability, repeatability, and operational efficiency of remote sensing-driven monitoring of ecological restoration in mining areas.

V. Conclusion

In this study, we explored the potential of combining Sentinel-1 and Sentinel-2 data for estimating AGB in the reclaimed mining area using four regression methods, including LMR, SVR, RF, and CB. The results indicate that combining Sentinel SAR and optical datasets enables accurate and stable estimation of AGB in reclaimed mining areas. Compared to single datasets, the complementary and combined vegetation structure information from Sentinel-1 and surface reflectance information from Sentinel-2 enhance model performance, mitigating the overestimation of low AGB values (<150 g/m^2) and the underestimation of high AGB values (>300 g/m^2). Among the four regression algorithms, SVR is the most robust in estimating the AGB of reclaimed mine sites. Despite the high accuracy of the models, the four machine learning algorithms do not have better temporal–spatial generalizability. Therefore, collecting ground truth data is essential when applying AGB models to new year or reclaimed sites to achieve high-accuracy prediction results. Finally, the most informative SAR metrics for AGB estimation are VV. The most important MSI metrics were associated with the VIS, red-edge, and SWIR spectral regions, mainly related to vegetation pigments and moisture. Precise and timely estimation of AGB can help to improve our understanding of the habitats of reclaimed mining areas and to plan and monitor land and biological resources, thereby minimizing the consequent risk of mismanagement of reclaimed mining areas.

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